4.4 DEPOSIT CREATION

Multiple Choice Questions

1990/CE/II/45
Which of the following will increase the money supply in an economy?

A. A citizen withdraws cash from a bank and keeps it in a safe.
B. A manufacturer repays a bank loan.
C. A bank grants a loan to a real estate developer.
D. A bank increases the amount of excess reserves it holds.

1991/CE/II/31
The withdrawal of money from banks causes

A. a decrease in the value of the maximum possible banking multiplier.
B. a decrease in the cash reserves held by banks.
C. an increase in the deposits created.
D. a decrease in the minimum legal reserve ratio.

1991/CE/II/32
Bank A has total deposits of $ 1 000 million. Its reserves are $ 240 million. Suppose the reserve ratio required by the government is 25 %. Which of the following will help Bank A fulfil reserve requirement?

A. Bank A increases its lending to customers by $ 40 million.
B. Bank A borrows $ 10 million from the inter-bank market.
C. Customers increase deposits with Bank A by $ 10 million.
D. Customers withdraw deposits from Bank A by $ 40 million.

1992/CE/II/34
Assuming that a 20% reserve ratio is required by law, when the banking system has excess reserves of $20 000, it can at most increase its deposits by

A. $80 000.
B. $100 000.
C. the amount of its total reserves.
D. the amount of its excess reserves.

1992/CE/II/35
The table below shows the total reserves, loans, and deposits of a bank:

Assets ($)		Liabilities ($)	
Reserves	200	Deposits	500
Loans	300		

From the table, we can say that

A. the legal minimum reserves are $200.
B. the bank holds excess reserves of $100.
C. the actual banking multiplier is 4.
D. the actual reserve ratio is 40%.

1992/CE/II/32
The following is a balance sheet of a banking system. The required reserve ratio is 20%

Assets ($)		Liabilities ($)	
Reserves	150	Deposits	1 000
Loans	850		
Total	1 000		1 000

Which of the following statements about the banking system is correct?

A. It has excess reserves.
B. It has to increase its reserves by $100.
C. It has to reduce loans by a total of $250.
D. The deposit multiplier is 4.

1994/CE/II/40
If the required reserve ratio for banks is 100%, which of the following is correct?

A. The money supply will not change if banks accept deposits from the public.
B. The total amount of bank loans must be equal to the total amount of deposits.
C. Banks have no chance to make profits.
D. The banking multiplier is zero.

1994/CE/II/42
Suppose the required minimum reserve ratio for banks is 25%. The table below shows the balance sheet of a certain bank.

Assets ($)		Liabilities ($)	
Reserves	600	Deposits	1 500
Loans	900		

Which of the following statements is correct?

A. The actual reserve ratio of the bank is 45%.
B. The excess reserves of the bank is $300.
C. The banking multiplier is 5.
D. The banking system can expand its loan to customers by a maximum of $900.

1995/CE/II/42
Suppose the legal reserve ratio is 100%. If a person deposits $500 cash into his current account,

(1) the currency in circulation will decrease by $500.
(2) the bank deposits will remain unchanged.
(3) the bank loans will increase by $500.
(4) the money supply will remain unchanged.

A. (1) and (2) only
B. (1) and (4) only
C. (2) and (3) only
D. (3) and (4) only

For Questions 35 and 36, refer to the following balance sheet of the banking system

Assets ($)		Liabilities ($)	
Cash reserve	250	Deposits	1 000
Loans	750		

Assume that legal minimum reserve ratio is 25 % and a person withdraws $ 40 from a bank.

1996/CE/II/35
How much would the cash reserve in the banking system fall short of the legal requirement?

A. $10
B. $20
C. $30
D. $40

1996/CE/II/36
What would be the maximum amount of deposits that the banking system can hold after the withdrawal?

A. $750
B. $840
C. $880
D. $960

1997/CE/II/33
Suppose the legal reserve ratio is 25%. If a person deposits $200 cash into his savings account, the maximum possible change in

A. bank loans is an increase of $800.
B. bank reserves is an increase of $200.
C. M1 is a decrease of $50.
D. M2 is an increase of $800.

1997/CE/II/34
If a $100 deposit of newly-printed money into a bank results in a $400 increase in the total deposit of the banking system, we can conclude that

A. the banking multiplier of the above deposit creation is 4.
B. the legal reserve ratio is 0.25.
C. the money supply increases by $500.
D. there is no cash leakage.

1998/CE/II/33
Study the following balance sheet of Bank B.

Assets ($)		Liabilities ($)	
Reserves	1 000	Deposits	3 000
Loans	2 000		

Suppose the required reserve ratio for banks is raised from 20% to 25%, what will be the decrease in Bank B's excess reserves?

A. $150
B. $250
C. $400
D. $600

1998/CE/II/34
Suppose the required reserve ratio for banks is 20%. The table below shows the balance sheet of the banking system.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	1 000
Loans	500		

Which of the following statements is correct?

A. The excess reserves of the banking system is $200.
B. The actual banking multiplier of the banking system is 5.
C. The banking system can expand its total deposits to $5 000.
D. The maximum amount of loans the banking system can make is $2 000

1998/CE/II/52
An amount of cash deposited into a bank will **NOT** lead to

A. an increase in bank loan made to the public
B. an increase in the total deposits of the banks.
C. an increase in the value of the maximum banking multiplier.
D. an increase in the cash reserves held by banks.

1999/CE/II/36
Refer to the following balance sheet of a banking system:

Assets ($)		Liabilities ($)	
Reserves	100	Deposits	400
Loans	300		

The legal reserve ratio is 20%. Suppose that all banks loan out their excess reserves and all loans are re-deposited into the baking system.

After the credit creation process is completed, the amount of reserves held by banks is ___________ and the amount of loans will increase by ___________.

A. less than $100 ...... more than $100
B. equal to $100 ...... $100
C. equal to $100 ...... more than $100
D. more than $100 ...... less than $100

1999/CE/II/38
Which of the following would reduce the amount of deposits created in the process of multiple credit creation?

A. An increase in the holding of cash by the public
B. An increase in the use of cheques by the public
C. An increase in the use of electronic money by the public
D. An increase in the bank loans made to the public

2000/CE/II/29
A bank accepts a deposit of $1 000 from a foreign country. Through the credit creation process, the amount of deposit of the whole banking system increases by $4 000.
Base on the above information, we can conclude that

A. The legal reserve ratio is 25%
B. Banks do not keep excess reserves.
C. The total money supply has increased by at least $4 000.
D. There is a cash leakage in the credit creation process.

2000/CE/II/30
As the use of the Octopus card (八達通卡) for public transport becomes much more popular,

A. The money supply will increase because the card can perform the functions of money.
B. The volume of the legal tender in circulation will drop.
C. The value of money will drop because the demand for cash will be smaller.
D. The maximum amount of deposits created by the banking system will decrease.

2001/CE/II/26
The following is the balance sheet of banking system.

Assets ($)		Liabilities ($)	
Reserves	200	Deposits	600
Loans	400		

Suppose the legal reserve ratio of the banking system is 20%. If all banks no longer hold excess reserves, the maximum increase in bank deposits is

A. $80
B. $160
C. $400
D. $1 000

Answer Questions 35 and 36 by referring to the following balance sheet of a banking system which has an excess reserve of $200.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	1 200
Loans	700		

2002/CE/II/35
The required reserve ratio is

A. 15%
B. 20%
C. 25%
D. 30%

2002/CE/II/36
Suppose someone has withdrawn $250 from the banking system. As a result, what would the maximum possible amount of bank deposits be?

A. $200
B. $600
C. $800
D. $1 000

2002/CE/II/37
Which of the following will reduce the deposit creation ability of the banking system?

(1) an increase in the use of credit cards by the public
(2) an increase in the amount of cash held by the public
(3) an increase in the legal reserve requirement for banks

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2002/CE/II/38
Assume the whole banking system does not hold excess reserves and the required reserve ratio is 20%. If a person withdraws $100 from a bank,

A. the amount of loans made by the banks to the public will fall by at most $400.
B. money supply will fall by at least $400.
C. the amount of deposits in the banking system will fall by at least $500.
D. the amount of cash reserve of the banks will fall by at most $80.

2003/CE/II/34
If the actual reserve is larger than the required reserve in a banking system, it follows that

A. there is cash leakage in the banking system.
B. some banks decide to hold excess reserves.
C. the demand for loans is insufficient.
D. the banking system can create more deposits.

Answer Questions 35 and 36 by referring to the following balance sheet of a banking system. Suppose the required reserve ratio is 20%.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	2 000
Loans	1 500		

2003/CE/II/35
Which of the following statements about the above banking system is correct?

A. The maximum banking multiplier is 4.
B. The maximum possible increase in loans will be the same as the maximum possible increase in deposits.
C. The actual reserve ratio is equal to the required reserve ratio.
D. The amount of reserves will decrease if all banks loan out their excess reserves.

2003/CE/II/36
If the government increases the required reserve ratio to 40%, which of the following statements about the above banking system is correct?

A. The amount of reserves will increase to $800.
B. The amount of deposits will not change.
C. The maximum amount of loans will be $750.
D. The maximum amount of deposits will be $1 600.

2004/CE/II/38
The following table shows the balance sheet of a bank in an economy:

Assets ($)		Liabilities ($)	
Cash reserves	500	Deposits	1 250
Loans and investments	750		

Suppose its customers withdraw $50 cash from the bank. Immediately after the withdrawal, the cash reserves and the deposits would be _______ and _______ respectively.

A. $450 ...... $1 200
B. $450 ...... $1 250
C. $500 ...... $1 200
D. $500 ...... $1 250

2004/CE/II/39
Refer to the following balance sheet of a banking system which has an excess reserve of $200.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	1 500
Loans	1 000		

Suppose $300 worth of deposits is withdrawn from the banking system. What will the maximum possible amount of bank deposits be?

A. $500
B. $1 000
C. $1 200
D. $6 000

2005/CE/II/37
Below is the balance sheet of a banking system.

Assets ($)		Liabilities ($)	
Cash reserves	500	Deposits	2 000
Loans	1 500		

Suppose the banking system has no excess reserves. After a withdrawal of $100 from the banks, the maximum possible amount of deposits of the banking system would be

A. $1 600
B. $1 700
C. $1 800
D. $1 900

2005/CE/II/50
The central bank of Country B increases the legal reserve ratio in order to discourage the overheating activities of lending and borrowing in the economy. Suppose the banking system has no excess reserves. The above measure adopted by the central bank would **NOT** reduce

A. the ability of deposit creation of the banking system.
B. the amount of reserves of the banking system.
C. the banking multiplier.
D. the money supply.

2006/CE/II/36
The table below shows the balance sheet of a banking system. The amount of its excess reserves is $800.

Assets ($)		Liabilities ($)	
Reserves	1 000	Deposits	2 000
Loans	1 000		

Which of the following statements about the banking system is correct?

A. The required reserve ratio is 50%.
B. The maximum amount of deposits is $8 000.
C. The maximum banking multiplier is 5.
D. The actual banking multiplier is 2.

2006/CE/II/37
During the period around Lunar new year every year, many people in Hong Kong withdraw cash from their deposit accounts with banks. Thus the amount of cash held by the public increases.

Which of the following is a definite consequence of the above situation?

A. The value of the maximum banking multiplier of the banking system of Hong Kong decreases.
B. The maximum possible amount of deposits of the banking system of Hong Kong decreases.
C. The amount of bank loans made to the public decreases.
D. The money supply of Hong Kong decreases.

2006/CE/II/38
Which of the following may reduce the size of the actual banking multiplier of a banking system?

A. an increase in the use of cheques by the public
B. an increase in the demand for bank loans
C. an increase in the risk of non-repayment of bank loans
D. a reduction in the required reserve ratio

2007/CE/II/36
The table below shows the balance sheet of a banking system, which has an excess reserve of $50.

Assets ($)		Liabilities ($)	
Reserves	250	Deposits	1 000
Loans	750		

Which of the following is correct?

A. The required reserve ratio is 25%.
B. The actual reserve ratio is 20%.
C. If $100 cash is deposited into the banking system, the total amount of deposits will be $1500.
D. If $100 is withdrawn from the banking system, the maximum possible total deposits will be $750.

2008/CE/II/38
Study the following balance sheet of a banking system which never holds excess reserves.

Assets ($)		Liabilities ($)	
Reserves	600	Deposits	3 000
Loans	2 400		

Suppose the public in this economy always holds the same amount of cash. If $200 is withdrawn from the bank deposits, the maximum possible amount of loans of the banking system would become _______.

A. $1 200
B. $1 600
C. $2 000
D. $2 200

2008/CE/II/39
Which of the following will increase the maximum banking multiplier?

A. Octopus cards are widely used.
B. The economy grows fast and investment opportunities flourish.
C. Banks keep more reserves to avoid the risk of bank runs.
D. Commercial banks are allowed to increase the proportion of customers' deposits that can be loaned out.

2009/CE/II/38
Suppose the actual banking multiplier is smaller than the maximum banking multiplier in a certain banking system. Which of the following would be a possible reason for this?

A. There is insufficient demand for bank loans.
B. The banking system is facing a reserve shortage.
C. The maximum possible amount of total deposits of the banking system will decrease.
D. The actual reserve ratio is smaller than the required reserve ratio.

2009/CE/II/39
Below is the balance sheet of a banking system which has $50 excess reserves.

Assets ($)		Liabilities ($)	
Reserves	350	Deposits	1 200
Loans	850		

Suppose all banks lend out their excess reserves and all loans will be redeposited into the banking system. After the deposit creation process is completed, the amount of reserves held by banks would be _________ and the amount of loans would be _________.

A. $300 ...... $1 400
B. $300 ...... $1 100
C. $350 ...... $1 400
D. $350 ...... $1 050

2010/CE/II/35
The maximum banking multiplier of a banking system will decrease when

A. the amount of excess reserves held by banks increases.
B. the required reserve ratio increases.
C. the amount of cash leakage from the banking system increases.
D. the demand for loans decreases.

2010/CE/II/36

Assets ($)		Liabilities ($)	
Reserves	2 500	Deposits	10 000
Loans	7 500		

The above table shows the balance sheet of a banking system which has an excess reserve of $5 00. Suppose the public withdraw $1 000 cash from the banking system. If the banks do not hold any excess reserves and there is no cash leakage, what will happen eventually after the process of deposit contraction is completed?

A. Bank deposits will fall by $5 000.
B. Bank loans will fall by $2 500.
C. The money supply will fall by $1 500.
D. Both B and C are correct.

2012/DSE/I/27
The table below shows the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	400	Deposits	1 000
Loans	600		

Suppose the required reserve ratio for the banking system is 25%. Which of the following statements is correct?

A. The excess reserves of the banking system are $200 million.
B. The actual banking multiplier of the banking system is 4.
C. The banking system can expand its total deposits to $1 600 million.
D. The maximum amount of loans the banking system can make is $1 000 million.

2013/DSE/I/26
Initially, a banking system has an excess reserve of$160 million and the public holds $200 million cash. The balance sheet of the banking system is as follows:

Assets ($ million)		Liabilities ($ million)	
Reserves	400	Deposits	1 200
Loans	800		

Suppose $100 million is withdrawn from the banking system and held as cash by the public. What will the money supply be if the banks lend out all excess reserves?

A. $1 500 million
B. $1 700 million
C. $1 800 million
D. $2 300 million

2014/DSE/I/30
Refer to the following balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	400	Deposits	1 200
Loans	800		

Suppose the required reserve ratio is 25%. If all banks loan out their excess reserves, which of the following statements about the above banking system is correct?

A. The amount of reserves held by the banks will decrease.
B. The maximum banking multiplier will increase.
C. The money supply will remain unchanged.
D. The maximum possible increase in loans will be the same as the maximum possible increase in deposits.

2016/DSE/I/31
Refer to the following balance sheet of a banking system. Its excess reserve is $125 million.

Assets ($ million)		Liabilities ($ million)	
Reserves	250	Deposits	500
Loans	250		

Which of the following statements is **INCORRECT?**

A. The required reserve ratio is 25%.
B. The maximum amount of loans is $750 million.
C. The banking system can expand its total deposit to $1 000 million.
D. The actual banking multiplier is 4.

2016/DSE/I/33

In a fractional reserve banking system, the actual deposit created is often below the maximum possible deposit created. The gap between these two amounts will be narrowed when

A. the popularity of electronic payment increases.
B. the central bank tightens the restrictions imposed on mortgage loans from banks.
C. the central bank buys bonds from the public.
D. the central bank reduces the legal reserve ratio for banks.

2017/DSE/I/31

The following table shows the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	300	Deposits	900
Loans	600		

Suppose the public does not hold cash and the required reserve ratio is reduced to 25%. After deposit creation, the amount of loans in the banking system is $750. Which of the following statements about the banking system is correct?

A. The money supply increases by $300.
B. The actual banking multiplier is 3.5.
C. The deposits increase by $600.
D. Bank reserves are $450.

2019/DSE/I/28

The following table shows the balance sheet of Country A's banking system. The money supply in Country A is $2 800 and all banks do not hold excess reserves.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	2 500
Loans	2 000		

Suppose the public deposits $200 cash into the banking system. Which of the following statements about the banking system is correct after the deposit creation process is completed?

A. The deposits would increase by $3 500.
B. The money supply would increase by $1 000.
C. The monetary base would increase by $200.
D. The loans will increase by $800.

2020/DSE/I/33

Which of the following will increase the actual banking multiplier?

A. The public tends to hold more cash.
B. The use of credit card becomes more popular.
C. The central bank increases the required reserve ratio.
D. The interest rate of bank deposit decreases.

2022/DSE/I/30

The balance sheet of a banking system is as follows.

Assets ($ million)		Liabilities ($ million)	
Reserves	500	Deposits	2 000
Loans	1 500		

The required reserve ratio is 20% and the public does not hold cash initially.

Suppose a customer withdraws $200 million from his bank account and holds it as cash. If all banks do not hold any excess reserves, the amount of money supply is ______________ after the process of credit creation/contraction has been completed.

A. $1 200 million
B. $1 400 million
C. $1 500 million
D. $1 700 million

2022/DSE/I/32

Suppose the central bank requires the commercial banks to keep all deposits as reserves. Which of the following statements is **INCORRECT**?

A. The monetary base will be equal to the money supply.
B. The required reserve ratio will be equal to one.
C. The maximum banking multiplier will be equal to zero.
D. The money creation ability of the banking system will be smaller than that of a fractional reserve banking system.

2023/DSE/I/34

Refer to the following balance sheet of a banking system which has excess reserves of $125 million.

Assets ($ million)		Liabilities ($ million)	
Reserves	500	Deposits	1 500
Loans	1 000		

Which of the following statements about the above banking system is correct?

A. The actual banking multiplier is 4.
B. If the banking system loans out all the excess reserves, the amount of reserves will decrease after the deposit creation process is completed.
C. If the banking system loans out all the excess reserves, the maximum possible increase in loans will be $500 million.
D. If the public withdraws $125 million from the banking system and holds it as cash, the amount of excess reserves will decrease to zero immediately.

2024/DSE/I/34

The table below shows the balance sheet of a banking system. Suppose there are excess reserves of \$20 million in the banking system and the public always holds a fixed amount of cash.

Assets (\$ million)		Liabilities (\$ million)	
Reserves	40	Deposits	100
Loans	60		

Suppose some banks lend out their excess reserves. After deposit creation, the loans in the banking system increase by \$20 million. Which of the following statements about the banking system is correct?

A. The reserves of the banking system become \$20 million.
B. The excess reserves of the banking system become \$16 million.
C. The banking system has a reserve shortage of \$8 million.
D. The amount of deposits in the banking system becomes \$200 million.

2025/DSE/I/34

Study the following balance sheet of a banking system.

Assets (\$)		Liabilities (\$)	
Reserves	250	Deposits	1 000
Loans	750		

Initially, the banking system has an excess reserve of \$50.
Suppose the public decides to reduce cash holdings by depositing \$100 cash into the banking system.
When the maximum deposit of the banking system is reached, which of the following will be its new balance sheet?

A.

Assets (\$)		Liabilities (\$)	
Reserves	300	Deposits	1 200
Loans	900		

B.

Assets (\$)		Liabilities (\$)	
Reserves	350	Deposits	1 400
Loans	1 050		

C.

Assets (\$)		Liabilities (\$)	
Reserves	300	Deposits	1 500
Loans	1 200		

D.

Assets (\$)		Liabilities (\$)	
Reserves	350	Deposits	1 750
Loans	1 400		

Short & Structured Questions

1990/CE/2/(d)(ii)
Suppose the banks in Hong Kong do not hold excess reserves and the legal minimum reserve ratio is 25%. Mr Chan withdraws HK$10 000 from his savings account with Bank A and keeps the money at home.

(I) Calculate the amount of reserves by which Bank A is falling short of the legal requirement. Suggest **ONE** possible way that Bank A can satisfy the reserve requirement again. (5 marks)

(II) Calculate the maximum possible change in the total deposits of the banking sector. (3 marks)

1991/CE/I/5(b)
Suppose there are only two banks, A and B, in an economy. They are subject to the same required cash reserve ratio. Bank B has no excess cash reserves. Their balance sheets are as follows:

Bank A

Assets ($)		Liabilities ($)	
Cash	50	Deposits	100
Loans	50		

Bank B

Assets ($)		Liabilities ($)	
Cash	200	Deposits	800
Loans	600		

(i) Calculate the required cash reserve ratio in the economy. (2 marks)

(ii) Calculate the amount of excess cash reserves of Bank A. (2 marks)

(iii) Now assume Bank A chooses not to hold excess cash reserves. Calculate the maximum possible increase in the deposits created by the whole banking system. (3 marks)

State **TWO** assumptions, other than those mentioned above, in arriving at your answer. (4 marks)

1992/CE/I/3(b)(ii)
In Hong Kong, Miss Wong puts $1 000 cash into The Hongkong and Shanghai Banking Corporation Limited (HSBC) as a demand deposit.

Explain **TWO** situations under which this $1 000 deposit will not lead to a further increase in deposits. (6 marks)

1993/CE/I/5(c)(ii)
Suppose Mr. Chan sells his flat to a friend in Canada and receives an oversea remittance which amounts to HK$3 million. He deposits the HK$3 million in his savings account with a bank in Hong Kong.

Given a reserve ratio of 25%, calculate the possible maximum increase in deposit (including the initial $3 million) of the local banking system. Show your working. (3 marks)

1994/CE/I/5
Suppose in a certain economy all the banks keep no excess reserves, and their balance sheet is as follows:

Assets ($)		Liabilities ($)	
Reserves	240	Deposits	1 200
Loans	960		

Suppose someone deposits $100 cash into his bank. What will be the maximum amount of deposits (including the initial $100) the banking system can create? Show your working. (4 marks)

1995/CE/I/10/(a)(iii)
The audience withdrew $M from their deposits with banks in the US to buy the concert tickets. Suppose the required reserve ratio of US banks was 20%, they never kept excess reserve and there was no cash drain. What would happen to the amount of the total deposit of banks in the US after the withdrawal and donation to Rwanda? Explain briefly the changing process. (8 marks)

1996/CE/I/11(c)
Suppose a newspaper publisher raises $5 million capital by issuing shares and deposits the full amount into the banks in Hong Kong. Assume that the required reserve ratio of the banks in Hong Kong is 25 % that they do not hold excess reserves and that there is no cash leakage.

Explain what would happen to the amount of the total deposit of the banks in Hong Kong after the shares issuing is complete if

(i) all the shares are purchased by local people who withdraw deposits from their banks in Hong Kong. (3 marks)

(ii) all the shares are purchased by foreigners who remit the money required from overseas to settle the purchase. (4 marks)

1997/CE/I/9(c)
Suppose the government builds a railway in northwest New Territories.
Suppose that the banks in Hong Kong have no excess reserves and the minimum reserve ratio is 25% and that $10 billion is borrowed overseas to construct the new railway.

Using the process of deposit creation, explain why the actual amount of bank deposits thus created in Hong Kong is smaller than the maximum amount of bank deposits can be created. (8 marks)

1998/CE/I/9(a)
Mr Wong received a remittance of HK$1 million from abroad and deposited the entire sum in a bank in Hong Kong. Later he used part of the money to buy shares and the rest to set up a restaurant.
Suppose the required reserve ratio of the banks in Hong Kong is 25% and the banks have no cash leakage and do not keep excess reserves.

Explain briefly the process of deposit creation of the local banking system caused by the injection of $1 million. Calculate the total amount of deposits created (including the initial $1 million). (8 marks)

1999/CE/I/6(a)
Suppose the legal reserve ratio for banks is 25 % and Bank A does not keep excess reserves. Aaron won an overseas scholarship and received a remittance of $10 000. He deposited it in Bank A. Bank A lent out all the excess reserves to GiGi. She then deposited the money into Bank B, but Bank B could not find any borrowers.

Calculate the amount of deposits thus created in the economy. (4 marks)

2000/CE/I/10(a)
The balance sheet of the banking system of Economy A is shown below. Suppose all banks do not hold excess reserves and the public holds $50 mn cash in hand.

Assets ($ mn)		Liabilities ($ mn)	
Cash reserves	200	Deposits	1 000
Loans	800		

Suppose the public now withdraws $10 million cash from the banks.

(i) Calculate the change in the total deposits of the banking system. (3 marks)

(ii) Calculate with explanation the change in the money supply. (4 marks)

2001/CE/I/10(b)
The balance sheet of a banking system is as follows:

Assets ($m)		Liabilities ($m)	
Reserves	250	Deposits	1 000
Loans	750		

(i) Given that the legal reserve ratio is 20%, calculate the actual reserve ratio and the amount of excess reserve of the banking system. (4 marks)

(ii) A depositor withdraws $40 m from the banking system and invests the money overseas. If all banks lend out their remaining excess reserves, what is the maximum possible amount of total deposits in the banking system? Explain your answer. (4 marks)

2002/CE/I/10(a)
The balance sheet of a banking system is as follows:

Assets ($mn)		Liabilities ($mn)	
Reserves	20	Deposits	100
Loans	80		

The public deposits $10 million of cash into the banking system.

(i) Suppose the banks do not hold excess reserves. Calculate the maximum possible change in the money supply. Explain your answer. (5 marks)

(ii) Give **TWO** possible reasons why the actual change in the money supply could be less than the maximum possible change. (4 marks)

2003/CE/I/11
(b) (ii) Using the process of deposit creation, explain how a decrease in demand for loans could affect the deposit creation ability of the banking system. (4 marks)

(c) Suppose the banking system has an excess reserve of $50 million and its balance sheet is as follows:

Assets ($m)		Liabilities ($m)	
Reserves	250	Deposits	1 000
Loans	750		

What is the maximum possible amount of total deposits in the banking system if no banks hold excess reserves? Show your workings. (4 marks)

2004/CE/I/6(b)
The balance sheet of Economy A's banking system is shown below. The public in this economy always holds $50 million in cash. Initially, the excess reserve of the banks is $100 million

Assets ($mn)		Liabilities ($mn)	
Reserves	500	Deposits	2 000
Loans	1 500		

Suppose $140 million worth of deposits is withdrawn from these banks and remitted overseas

Explain briefly the process of deposit contraction of the above withdrawal. (4 marks)

2005/CE/I/5

Study the following balance sheet of a banking system:

Assets ($mn)		Liabilities ($mn)	
Reserves	120	Deposits	1 000
Loans	880		

Suppose the legal reserve ratio is 10%

(a) Calculate the amount of excess reserve of the banking system. (2 marks)

(b) Suppose all banks lend out all their excess reserves. Calculate the maximum possible amount of total deposits in the banking system. (2 marks)

2006/CE/I/6

The following is the balance sheet of a banking system.

Assets ($)		Liabilities ($)	
Reserves	500	Deposits	1 500
Loans	1 000		

(a) Suppose the excess reserve of the banks is $350. What is the required reserve ratio?

The required reserve ratio is ____________ %. (1 mark)

(b) Suppose there is a cash withdrawal of $350 from the banks. Explain whether the banks still have excess reserves immediately after the withdrawal. (3 marks)

2007/CE/I/7

Study the following balance sheet of a banking system:

Assets ($)		Liabilities ($)	
Reserves	1 700	Deposits	7 500
Loans	5 800		

Suppose the legal reserve ratio is 20%.

(a) The excess reserve of the banking system is $ ____________. (1 mark)

(b) Suppose Mr Chan withdraws $100 from the banking system, of which 70 % is held as cash and the rest is remitted overseas.

(i) The immediate change in total deposits of the banking system is ______________. (1 mark)

(ii) Therefore, the immediate change in the overall amount of the money supply is ______________. (2 marks)

2008/CE/I/10

The government may reduce the money supply to cool down the overheated economy. The balance sheet of the banking system is as follows:

Assets ($ million)		Liabilities ($ million)	
Reserves	1 000	Deposits	5 000
Loans	4 000		

Suppose the public in this economy always holds the same amount of cash and the banking system never holds excess reserves. The central bank increases the minimum reserve ratio of the banks by 5%.

(a) Explain briefly the process of deposit contraction resulting from the above change in. the reserve ratio. (4 marks)

(b) Calculate and explain the change in money supply in the economy resulting from the above deposit contraction. (5 marks)

2009/CE/I/6

Study the following balance sheet of a banking system:

Assets ($)		Liabilities ($)	
Reserves	300	Deposits	1 000
Loans	700		

Suppose the legal reserve ratio is 20%.

(a) Calculate the excess reserve of the banking system. (2 marks)

(b) Suppose all the excess reserve is loaned out. Calculate the maximum possible amount of total deposits in the banking system. (2 marks)

2010/CE/I/11(b)

The 2008 financial tsunami had a huge impact on Economy A.

Suppose the balance sheet of the banking system was as follows:

Assets ($ million)		Liabilities ($ million)	
Reserves	2 000	Deposits	8 000
Loans	6 000		

During the credit crisis, the banks increased the reserve ratio to 40% for safety purposes. This resulted in a change in the bank deposits.

(i) Calculate the maximum possible change in the bank deposits. (3 marks)

(ii) Explain briefly the process of deposit contraction resulting from the above change in the reserve ratio. (4 marks)

(Note: Although the topic of deposit creation was included in the HKAL syllabus, most questions are out of the current DSE syllabus, and the focus of the rest was mainly on the concepts of monetary base and monetary policy.)

1997/AL/II/9
Consider the following balance sheet of a banking system.

Assets ($)		Liabilities ($)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Assumes that the public does not hold any cash.

(a) Suppose banks are fully loaned up. What is the required reserve ratio? (1 mark)

(b) Suppose the maximum demand for loan is $4 500. Since the banking system has only extended $3 000 in loan, there is still an excess demand for loan of $1 500. If the government changes the required ratio to 10%, what will be the money supply? Explain your answer. (5 marks)

(c) Based on your answer in (b), comment on whether a government can completely control the money supply. (4 marks)

1998/AL/II/9
Consider the following balance sheet of a banking system.

Assets ($)		Liabilities ($)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Suppose the public holds $500 cash and the banks are fully loaned up.

(a) What is the money supply? (2 marks)

(b) Suppose the public only wants to hold $400 cash and deposits any excess cash into the banking system. Explain the impact of this on the balance sheet of the banking system and on the money supply in the economy in each of the following two situations.

(i) Banks cannot lead out any amount of the excess reserves. (3 marks)

(ii) Banks can lead out all the excess reserves. (3 marks)

(c) Using your results in (a) and (b), explain the impact on the money supply of a reduction in the public's desire to hold cash. (2 marks)

2017/DSE/II/8
The following table shows the balance sheet of a banking system which has excess reserves of $250.

Assets ($ million)		Liabilities ($ million)	
Reserves	1 000	Deposits	3 000
Loans	2 000		

(a) Find the required reserve ratio. (1 mark)

(b) Suppose the banks lend out all excess reserves and the public does not hold cash, Calculate the change in deposits after credit creation, Show your working, (2 marks)

2019/DSE/II/7

The following is the balance sheet of a banking system.

Assets ($million)		Liabilities ($million)	
Reserves	300	Deposits	1 000
Loans	700		

Suppose the legal reserve ratio is 20% and the public holds $200 million cash.

(a) Calculate the excess reserves of the banking system. (1 mark)

(b) Suppose the public no longer holds cash and all banks lend out all excess reserves. Calculate the maximum possible change in money supply. Show your workings. (4 marks)

2021/DSE/II/6

The following table shows the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	600	Deposits	1 500
Loans	1 200		

Initially the banks hold excess reserves of $300 million and the public always holds $100 million cash.

(a) Find the required reserve ratio. (1 mark)

(b) Suppose the required reserve ratio is reduced by 5% and banks lend out all excess reserves. Calculate the new money supply after the credit creation/contraction. Show your workings. (3 marks)

2022/DSE/IB/8

The following table shows the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

The public initially holds $1 000 million cash and the banks are fully loaned up.

(a) Calculate the required reserve ratio. (1 mark)

(b) Suppose the public only wants to hold $400 million cash and deposits the remaining cash into the banks. Calculate the maximum change in money supply. Show your workings. (3 marks)

2023/DSE/II/10i, ii

In 2022, the Hong Kong Financial Secretary announced the issuance of Silver Bonds for citizens aged 60 or above to subscribe. The nominal interest rate of the Silver Bonds is equal to the actual inflation rate in Hong Kong, subject to a minimum of 4%.

(c) Suppose a large number of senior citizens withdraw money from their bank accounts to purchase Silver Bonds.

(i) Describe the process of deposit creation/con traction which may result from the above withdrawal. (3 marks)

(ii) Explain with **ONE** reason why the above withdrawal does not necessarily lead to deposit creation/contraction in reality. (2 marks)

2024/DSE/II/8

Suppose in an economy, the public increases their cash holding by withdrawing bank deposits. Immediately after the withdrawal, the banking system of the economy faces a reserve shortage of $100 million.

The following table shows the balance sheet of the banking system immediately after the withdrawal.

Assets ($ million)		Liabilities ($ million)	
Reserves	600	Deposits	3 500
Loans	2 900		

(a) Find the required reserve ratio. (1 mark)

(b) Suppose the required reserve ratio is reduced by 5 percentage points. Calculate the maximum possible change in deposits after deposit creation/contraction. Show your workings. (3 marks)